A Structural Free Energy on the Solution Manifold

Coherence Diagnostics, Critical Coherence Index,

and Regime Classification in Nonlinear Field Systems

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Abstract

Many nonlinear field theories admit a large space of mathematically valid solutions, yet only a small subset tends to appear as physically persistent structures. This paper develops a structural diagnostic framework for analysing the stability, coherence, and dynamical regimes of field configurations without modifying the underlying dynamics.

We introduce an effective structural free-energy-like functional defined on the solution manifold of the dynamical equations and a dimensionless instability measure, the *Critical Coherence Index* (CCI), which plays a role analogous to Reynolds-type numbers in fluid dynamics. The CCI combines five structural diagnostics — dynamical coherence, conservation integrity, correlation structure, non-equilibrium activity, and constraint integrity — into a single dimensionless ratio separating ordered, critical, and chaotic dynamical regimes.

The framework connects structural diagnostics, information-theoretic measures, and non-equilibrium dynamics, and provides a basis for numerical investigation of structural selection phenomena in nonlinear field theories. It extends and complements the structural selection programme of [1, 2, 3].

Contents

1 Introduction **2**

2 Structural Diagnostics **2**

2.1 Dynamical Coherence Γ_{coh} . . . 2

2.2 Conservation Integrity Γ_{cons} . . . 2

2.3 Correlation Structure Γ_{corr} . . . 3

2.4 Non-equilibrium Activity Γ_{prod} . . . 3

2.5 Constraint Integrity Γ_{int} . . . 3

3 Structural Free Energy **4**

4 Structural Imbalance **4**

5 Critical Coherence Index **4**

6 Regime Classification **5**

7 Structural Reynolds Number **5**

8 Connections to Known Physics **5**

9 Interpretation of the Structural Diagnostics **5**

10 Numerical Implementation 6

11 Conclusion 6

1 Introduction

Nonlinear field equations typically admit a wide variety of solutions, including:

- homogeneous vacuum configurations,
- topological solitons and domain walls,
- localised wave packets,
- chaotic or turbulent states.

A central question is:

Why do certain configurations persist while others rapidly decay or become chaotic?

Examples include soliton stability in nonlinear wave equations, the onset of turbulence in fluid-like field systems, and pattern formation in reaction-diffusion and scalar field models. In each case, the dynamical equations alone do not select a preferred configuration — a second principle is needed.

Rather than modifying the fundamental dynamics, we consider a second layer of analysis defined on the *solution manifold* of the equations of motion:

$$\mathcal{M}_S = \{\Phi \mid \delta S[\Phi] = 0\}. \tag{1}$$

Within this manifold, structural diagnostics can be used to quantify properties such as stability, coherence, correlation, and dynamical activity — and to classify configurations into ordered, critical, and chaotic regimes through a single dimensionless index.

Figure 1 gives a schematic overview of the framework.

2 Structural Diagnostics

We introduce a set of normalised structural diagnostics $\Gamma_i \in [0, 1]$ measuring distinct structural aspects of a field configuration.

2.1 Dynamical Coherence Γ_{coh}

Measures how smoothly and coherently the system evolves. Possible proxies include gradient regularity or Euler–Lagrange residual norms:

$$\Gamma_{\text{coh}} \sim 1 - \frac{\|R[\Phi]\|^2}{\Lambda_R + \|R[\Phi]\|^2}, \qquad R_i \equiv \frac{\delta S}{\delta \Phi_i}. \tag{2}$$

2.2 Conservation Integrity Γ_{cons}

Quantifies how well conservation laws are respected:

$$\Gamma_{\text{cons}} \sim 1 - \frac{\|\nabla_\mu T^{\mu\nu}\|^2}{\Lambda_T + \|\nabla_\mu T^{\mu\nu}\|^2}. \tag{3}$$

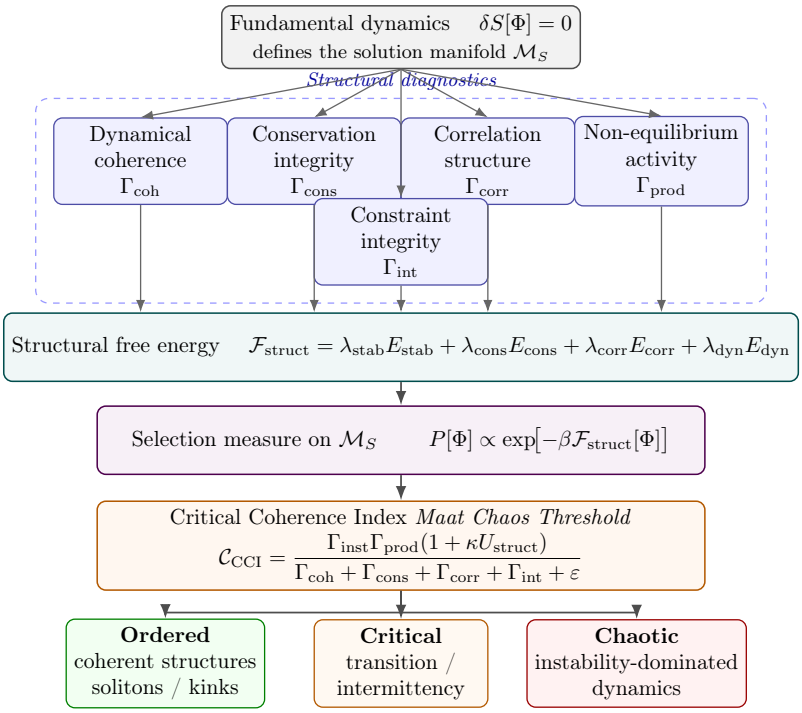

Figure 1: Conceptual structure of the framework. Fundamental dynamics defines the solution manifold; five structural diagnostics define an effective structural free-energy-like functional; the resulting selection measure classifies configurations into ordered, critical, and chaotic regimes through the Critical Coherence Index (CCI).

2.3 Correlation Structure Γ_{corr}

Measures connectivity or spatial correlations in the field configuration. Examples include mutual information, correlation length, or gradient coherence:

$$\Gamma_{\text{corr}} \sim \frac{I(\phi_i : \phi_{i+1})}{I_0 + I(\phi_i : \phi_{i+1})}. \tag{4}$$

In numerical implementations, I is approximated using local mutual information estimators (e.g. histogram-based) or spatial correlation length proxies; the exact estimator should be chosen consistently with the lattice geometry and periodicity of the model.

2.4 Non-equilibrium Activity Γ_{prod}

Measures the presence of dynamical activity. Typical proxies capture kinetic and gradient contributions:

$$\Gamma_{\text{prod}} = \frac{\langle \pi^2 \rangle + \alpha \langle |\nabla \phi|^2 \rangle}{S_0 + \langle \pi^2 \rangle + \alpha \langle |\nabla \phi|^2 \rangle}. \tag{5}$$

2.5 Constraint Integrity Γ_{int}

Encodes structural constraints such as boundary conditions, absence of forbidden field values, and anomaly cancellation:

$$\Gamma_{\text{int}} \sim 1 - \frac{\langle \max(0, |\phi| - \phi_{\max})^2 \rangle}{\Lambda_b + \langle \max(0, |\phi| - \phi_{\max})^2 \rangle}. \tag{6}$$

3 Structural Free Energy

We define an effective structural free-energy-like functional on the solution manifold:

$$\boxed{\mathcal{F}_{\text{struct}}[\Phi] = \lambda_{\text{stab}}\, E_{\text{stab}} + \lambda_{\text{cons}}\, E_{\text{cons}} + \lambda_{\text{corr}}\, E_{\text{corr}} + \lambda_{\text{dyn}}\, E_{\text{dyn}}} \tag{7}$$

where the $\lambda_i > 0$ weight contributions from each structural sector. This functional plays a role analogous to a free energy in statistical physics, though it is not a thermodynamic free energy in the strict sense.

Configurations on $\mathcal{M}_S$ are weighted according to a Boltzmann-type selection measure:

$$P[\Phi] = \frac{1}{Z}\exp\bigl(-\beta\,\mathcal{F}_{\text{struct}}[\Phi]\bigr), \qquad Z = \int_{\mathcal{M}_S} \mathcal{D}\Phi\; e^{-\beta\mathcal{F}_{\text{struct}}[\Phi]}. \tag{8}$$

Thus the dynamics determines the space of possible configurations, while the structural free energy determines which configurations are statistically preferred. This separation between dynamical admissibility and structural preference is the central conceptual contribution of the framework.

4 Structural Imbalance

Complex systems often become unstable not because any single diagnostic is extreme, but because the structural sectors become strongly *imbalanced*. We quantify this sectoral imbalance as the standard deviation of the five diagnostics:

$$U_{\text{struct}} = \sqrt{\frac{1}{5}\sum_{i=1}^{5}(\Gamma_i - \mu)^2}, \qquad \mu = \frac{1}{5}\sum_{i=1}^{5}\Gamma_i, \tag{9}$$

where the sum runs over $i \in \{\Gamma_{\text{coh}}, \Gamma_{\text{cons}}, \Gamma_{\text{prod}}, \Gamma_{\text{corr}}, \Gamma_{\text{int}}\}$. A large U_{struct} indicates that one sector dominates or one sector collapses — both are precursors of instability.

This connects to the MAAT-framework insight [1]: *not high force, but unbalanced force produces chaos.*

5 Critical Coherence Index

To characterise dynamical regimes we define the **Critical Coherence Index** (CCI):

$$\boxed{\mathcal{C}_{\text{CCI}} = \frac{\Gamma_{\text{inst}}\,\Gamma_{\text{prod}}\,(1 + \kappa\, U_{\text{struct}})}{\Gamma_{\text{coh}} + \Gamma_{\text{cons}} + \Gamma_{\text{corr}} + \Gamma_{\text{int}} + \varepsilon}} \tag{10}$$

where Γ_{inst} is an instability rate proxy (e.g. estimated from growth rates of local perturbations or Lyapunov-type indicators), $\kappa \geq 0$ controls sensitivity to sectoral imbalance, and $\varepsilon > 0$ prevents singular behaviour.

The CCI has the structure of a Reynolds number: *destabilising numerator* divided by *stabilising denominator*. When the numerator (instability $\times$ activity $\times$ imbalance) exceeds the denominator (sum of coherence-maintaining sectors), the system transitions toward chaotic dynamics.

6 Regime Classification

The CCI separates three dynamical regimes:

$$\mathcal{C}_{\mathrm{CCI}} < \tau_1 \quad \Rightarrow \quad \text{ordered regime (coherent structures, solitons, kinks)}, \tag{11}$$

$$\tau_1 \leq \mathcal{C}_{\mathrm{CCI}} < \tau_2 \quad \Rightarrow \quad \text{critical regime (transition, intermittency)}, \tag{12}$$

$$\mathcal{C}_{\mathrm{CCI}} \geq \tau_2 \quad \Rightarrow \quad \text{chaotic regime (instability-dominated dynamics)}. \tag{13}$$

The thresholds τ_1, τ_2 are system-dependent and to be determined from calibration against known transitions. This regime structure parallels the role of the Reynolds number in fluid turbulence, where a single dimensionless ratio characterises the laminar–turbulent transition.

7 Structural Reynolds Number

In the limit $\kappa \to 0$ (no imbalance sensitivity), the CCI reduces to a *Structural Reynolds Number*:

$$\boxed{\mathcal{R}_{\mathrm{struct}} = \frac{\Gamma_{\mathrm{inst}}\,\Gamma_{\mathrm{prod}}}{\Gamma_{\mathrm{coh}} + \Gamma_{\mathrm{cons}} + \Gamma_{\mathrm{corr}} + \Gamma_{\mathrm{int}}}} \tag{14}$$

This dimensionless ratio measures the balance between destabilising and stabilising structural processes, and provides the simplest diagnostic for regime classification.

8 Connections to Known Physics

The framework connects naturally to several established areas.

Turbulence. Reynolds-type numbers describe the transition between laminar and turbulent flow. The CCI is the field-theoretic analogue: a single dimensionless ratio characterises when coherent structure gives way to chaotic dynamics.

Non-equilibrium thermodynamics. Prigogine's theory of dissipative structures [6] shows that complex patterns emerge only within a window of regulated dynamical activity — neither frozen nor chaotic. The activity window in the structural free energy formalises this insight quantitatively.

Phase transitions. The CCI can act as a dynamical order parameter for transitions between coherent and chaotic regimes, analogous to an order parameter in a statistical mechanical phase transition. The numerical evidence in [3] for a selection transition in the activity target parameter $s_\star$ provides a first example of such a structural transition.

Information theory. Correlation measures such as mutual information [7] provide a natural description of structural connectivity (Γ_{corr}), connecting the framework to information-theoretic approaches to complexity.

9 Interpretation of the Structural Diagnostics

It is important to emphasise that the diagnostics introduced in this framework do not determine whether a configuration is physically realised in nature. They instead provide a structural classification of dynamical regimes within the solution manifold of a given model.

The fundamental dynamics of the system remain fully determined by the field equations $\delta S[\Phi] = 0$, which define the space of admissible configurations. The structural diagnostics and the resulting Critical Coherence Index operate on this solution manifold as *descriptive measures* of the dynamical state of a configuration — not as criteria for physical validity.

Consequently, extreme values of the diagnostics do not imply physical impossibility. Low coherence, strong dynamical activity, or large structural imbalance may simply correspond to

highly dynamical regimes such as turbulence, intermittency, or strongly nonlinear evolution. In many physical systems — for example plasmas, turbulent fluids, or cosmological structure formation — such regimes are not pathological but represent natural dynamical states.

In this sense, the Critical Coherence Index should be interpreted analogously to dimensionless instability parameters such as the Reynolds number in fluid dynamics. A high value of the index does not signal an unphysical configuration, but rather indicates that destabilising processes dominate over coherence-preserving mechanisms, leading the system into a chaotic or instability-driven regime.

The purpose of the diagnostics is therefore not to determine physical validity, but to provide a systematic way to classify and compare dynamical regimes within nonlinear field systems. This structural perspective allows ordered, critical, and chaotic dynamics to be characterised within a unified diagnostic framework while leaving the underlying physical dynamics unchanged.

10 Numerical Implementation

For lattice field simulations, one practical implementation is:

$$\Gamma_{\text{inst}} = \frac{\langle \pi^2 \rangle}{\Lambda_\pi + \langle \pi^2 \rangle}, \tag{15}$$

$$\Gamma_{\text{prod}} = \frac{\langle \pi^2 \rangle + \alpha \langle |\nabla \phi|^2 \rangle}{S_0 + \langle \pi^2 \rangle + \alpha \langle |\nabla \phi|^2 \rangle}, \tag{16}$$

$$\Gamma_{\text{corr}} = \frac{I(\phi_i : \phi_{i+1})}{I_0 + I(\phi_i : \phi_{i+1})}, \tag{17}$$

where I denotes mutual information between neighbouring sites, and $\langle \cdot \rangle$ denotes spatial mean. All quantities are bounded in $[0, 1)$ by construction.

The companion code at [5] implements these diagnostics for 1D and 2D ϕ^4 field theories and is available for direct use and extension.

11 Conclusion

We have introduced a structural framework for classifying dynamical regimes in nonlinear field theories. Its three central elements are:

1. an effective *structural free-energy-like functional* $\mathcal{F}_{\text{struct}}[\Phi]$ on the solution manifold, weighting configurations by their structural coherence,

2. a *structural imbalance measure* U_{struct} quantifying sectoral inequality among the five diagnostics, and

3. a *Critical Coherence Index* $\mathcal{C}_{\text{CCI}}$ separating ordered, critical, and chaotic regimes through a single dimensionless ratio.

Together these tools provide a quantitative language for analysing stability, structure formation, and the onset of chaos in nonlinear dynamical systems, grounded in the structural selection programme of the companion papers [1, 2, 3].

The determination of the thresholds τ_1, τ_2 and the imbalance sensitivity κ from first principles remains an open problem, as does the extension of the framework to quantum field theories and cosmological settings. Future work will test the proposed index in additional nonlinear field models — including sine-Gordon, three-dimensional systems, and driven-dissipative extensions — and investigate whether universal threshold values exist across dimensions and model classes.

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